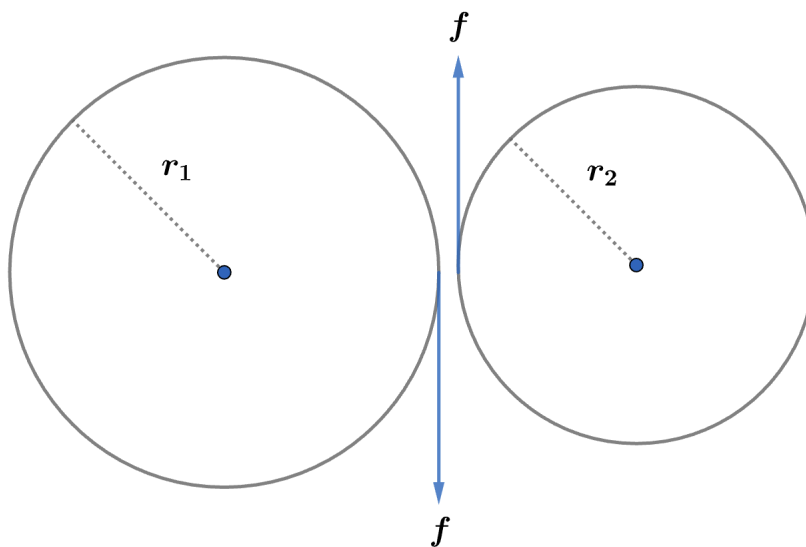


2009 F=ma Exam: Problem 25

Kevin S. Huang



Since both discs come to rest, they go to rest at the same time. From angular kinematics,

$$\omega - \alpha t = 0$$

$$t = \frac{\omega}{\alpha}$$

so the quantity ω/α is the same for both disks:

$$\frac{\omega_1}{\alpha_1} = \frac{\omega_2}{\alpha_2}$$

Let's rewrite α in terms of r . We have

$$\alpha = \frac{\tau}{I} \propto \frac{r}{mr^2} \propto \frac{r}{r^4} = \frac{1}{r^3}$$

using the fact that $\tau \propto r$ as the same force (friction) acts on both disks and also $m \propto r^2$ as both disks have the same density and thickness. Substituting into the earlier equation,

$$\omega_1 r_1^3 = \omega_2 r_2^3$$

so the answer is D.